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# Caracterización de la continuidad de una aplicación lineal

**Teorema 1.** *Sean $X$ e $Y$ dos espacios vectoriales normados y sea $T \colon X \to Y$ una aplicación lineal, entonces, son equivalentes:*

(i) *$T$ es uniformemente continua.*

(ii) *$T$ es continua.*

(iii) *$T$ es continua en $0 \in X$.*

(iv) *$\exists x_0 \in X : T$ es continuo en $x_0$.*

(v) *$\exists C > 0 : \forall x \in X : \|T(x)\|_Y \leqslant C \|x\|_X$.*

## Referenciado en

- `teo-hahn-banach-ii`
- `teo-esp-normado-separacion-punto-subespacio-cerrado`
- `prop-funcional-lineal-continuo-prod-interno`
- `teo-esp-reflexivo-iff-dual-reflexivo`
- `lem-separacion-punto-conjunto-convexo-abierto`
- `prop-apl-adjunta-lineal-continua-norma`

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